

I. RMQ .. But?

time limit per test: 2 seconds
memory limit per test: 256 megabytes
input: rmq.in

Given an array a of n integers $(a_1, a_2, a_3, ..., a_n)$.

For a given l and r , lets define a F function that calculates the maximum integer in the range starting at l and ending at r , in other words $F(l, r) = \max(a_l, a_{l+1}, ..., a_{r-1}, a_r)$.

You need to perform q queries of the following types:

type 1) 1 l r : For all $l \leq i \leq r$, set $a_i := F(l, r) - a_i$.

type 2) 2 l r : Print the value of $F(l, r)$.

Input

First line of the input contains an integer t , the number of test cases.

Each test case begins with two integers n, q ($1 \leq n, q \leq 10^5$) denoting size of array and number of queries respectively.

The second line of each case contains n integers a_i ($1 \leq a_i \leq 10^5$).

q lines follow, each containing three integers $type, l, r$ ($type \in \{1, 2\}, 1 \leq l \leq r \leq n$).

Output

For each test case answer all queries of type 2 with a line containing a single integer: the value of $F(l, r)$.

Example

input	Copy
1 5 6 1 2 3 4 5 1 1 3 2 1 1 2 2 2 2 3 3 1 1 5 2 1 5	
output	Copy
2 1 0 5	

Note

Initially, the array is $[1, 2, 3, 4, 5]$, after the first query the array becomes $[2, 1, 0, 4, 5]$. After the fifth query, the array becomes $[3, 4, 5, 1, 0]$.

ACPC Scientific Committee Archive

Private

Participant



About Group

https://www.youtube.com/@MoamenAhmed98/play

Group website

Group Contests

- ECPC Qualifications 2025 Day 1 (Mirror)
- TCPC 2025
- ACPC Girls 2025
- ACPC Girls 2020
- HPV-CPC 2021

- Tunisian Collegiate Programming Contest (2017)
- Azhar Collegiate Programming Contest 2019

- The 2018 Menofia University Collegiate Programming Contest
- ICPC International Collegiate Programming Contest, Assiut Collegiate Programming Contest (2019)

- International Collegiate Programming Contest, GUC Collegiate Programming Contest (2019)

- International Collegiate Programming Contest, Aleppo University Collegiate Programming Contest(2018)

- The 2025 ICPC Egyptian Collegiate Programming Contest for Teens(Online Qualification)

- ECPC Qualifications 2021 - Day 4

- ACPC Kickoff 2025

- International Collegiate Programming Contest, Al-Baath, Tishreen, Aleppo and Damascus Collegiate Programming Contest (2021)

- The 2020 Egyptian Collegiate Programming Qualification Contest Day 2

- OCPC 2025 (Qualification)

- OCPC 2025

- ACPC 2024 (Mirror)

- LCPC 2024 (Mirror)

- JCPC 2024 (Mirror)

- EthCPC (Mirror)

- PCPC 2024

- LyCPC 2024

- AOCPC 2024

- AOCPC 2023

- QCPC 2024 (Mirror)

- TCPC 2024 (Mirror)

- AICPC 2023

- SCPC 2024 (Mirror)